PROCESS OR MATERIALS REVIEW REQUEST FORM

SNF Stanford Nanofabrication Facility

Request Title: Coating Process with Nanotube Ink

Requestor: Paul Comita and Ludwig Galambos/Nanosense **Date:** Dec 11, 2018

Non-SNF staff, please complete the following:

Badger ID: pcomita50@gmail.com

Company name/PI: Nanosense/Paul Rhodes

Send completed form to: snf-promcommittee@lists.stanford.edu

1. **The chemical or material. NanoIntegris SKU#1101 IsoNanotubes-S 99% Aqueous 1.0mg The Storage Group Identifier is G (not intrinsically reactive, flammable or combustible) Health Risk - 1 Fire Risk - 1 Reactivity - 0**

2. **Vendor/manufacturer info:** NanoIntegris Technologies
   3765 La Verendrye
   Boisbrian, QC, J7H 1R8
   Canada
   650-365-3120

3. **Reason for request:** The CNT ink will be used to coat patterned wafers. The process has been developed by Nanosense, and this step is detailed below.

4. **Process Flow:** CNT Coating Process for patterned 4" wafers. These wafers will have gold/chrome metal, so will be classified as gold contaminated. The patterning will be done with tools at SNF (Heidelberg/AJA/Webflexsolv), using standard processes. Clean the wafer - this will be done post-patterning
   APTES coating will be done at an outside laboratory(Integrated Surface Technologies)
   Soak in CNT ink (undiluted, centrifuged) for 45 min at Webflexcorr using pyrex dishes.
   Allow excess CNT ink to drip/flow off and allow wafers to slowly air dry at Webflexcorr.
   The follow-on procedures and tools are:
   1. LOL-3612 coating (Headway/SVGCoat), exposure (Heidelberg), develop (SVGDEV)
   2. O2 plasma etch (PTOx or Oxford RIE)
   3. Remove resist (WBFlexsolv)
   4. bake (likely BlueM Oven)
   5. Spot check resistance and follow-on IV measurements at probe station.

5. **Storage:** Will you be storing your chemical/material at SNF? Yes, storage Group G.

6. Amount and form. How much will you bring in? 1mg in liquid form. Do you need to mix it to use it? It may need to be centrifuged or shaken if it has settled.

7. **Disposal**: How will you dispose of any waste or excess chemical or material?

According to the MSDS, if this product becomes a waste, it would not be expected to meet the criteria of hazardous waste. However, I would collect the excess, used material and would label and dispose of as required by SNF.

To be completed by the PROM COMMITTEE:

Special handling requirements:

On-going approval or staff interaction requirements:

Approved By/Date: